GUBELOG

Technical datasheet

GUBELOG-01 · GUBELOG-01R · GUBELOG-03 · GUBELOG-03R · Revision 0014 · November 2026





GUBELLINI s.a.s. di Diego Gubellini & C. – Via Euridia Bergianti 10B, 40059 Medicina (BO) Italy
P.IVA / VAT IT03466001207 · info@gubellinielectronics.com · Tel. +39 347 9200208 · WhatsApp +39 380 1978288
www.gubellinielectronics.com

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1 Overview

1.1 Description

GUBELOG is a professional multifunction data logger, designed to acquire physical signals and on-board data accurately and reliably in motorsport, automotive, motorcycle, laboratory and industrial monitoring applications. A compact, watertight (IP67) housing integrates a GPS/GNSS receiver, a 6-axis inertial platform, analog and digital inputs, a 16-channel CAN bus and two digital outputs for driving relays.

Data is recorded on microSD and downloaded via USB-C or by reading the microSD on the PC. Configuration, download and real-time monitoring are done with the free **GUBELLINI DataStudio** software; data analysis with the free **GUBELLINI DataAnalysis** software (chap. 5).

1.2 Main features

- **38 inputs** recorded simultaneously: 4 GPS, 6 IMU, 4 digital, 8 analog, 16 CAN.
- **Sampling rate** programmable per channel from **0.01 Hz to 200 Hz**: suitable both for long-term monitoring and for dynamic analysis.
- Integrated **GPS/GNSS** (1 Hz on GUBELOG-01, 10 Hz on GUBELOG-03) and **6-axis IMU** (accelerometers ±30 g, gyroscopes ±2000 °/s).
- **8 analog inputs**, 12-bit: 4 voltage inputs 0-5 V and 4 current inputs 0/4-20 mA.
- **4 digital / frequency inputs** from 1 Hz to 20 kHz (engine speed, wheel speed, flow meters).
- **CAN 2.0B** up to 16 signals: OBD-II, SAE J1939, ISOBUS (ISO 11783), CANopen (EN 50325-4) or custom signals, also from DBC files.
- **CAN Bridge**: transmits the values of the physical inputs on the CAN bus in real time.
- **2 digital outputs** driven by programmable alarms, to control relays, pumps, solenoid valves.
- Integrated **WiFi** with cloud upload (GUBELOG-01 / 01R only).
- **IP67**, from **-20 °C to +85 °C**, 26-way **JAE MX23A** automotive connector.

2 Versions

2.1 Model comparison

GUBELOG is available in four versions. **GUBELOG-01** is designed for remote monitoring and integrates the WiFi module, which connects to the local network and communicates with the cloud; **GUBELOG-03** is optimised for performance in motorsport, automotive and sports applications. Both are available in the lightweight version and in the rugged **R** version.

	GUBELOG-01	**GUBELOG-01R**	**GUBELOG-03**	**GUBELOG-03R**
Order code	GE-GL-0001	GE-GL-0001R	GE-GL-0003	GE-GL-0003R
Typical application	Remote monitoring, laboratory, industry		Motorsport, automotive, sports applications	
GPS/GNSS	1 Hz, antenna integrated on the board		10 Hz, external antenna	
WiFi module	Yes (WIZnet WizFi360)		No	
Housing	PA12 + 35% GF	Extruded anodised aluminium	PA12 + 35% GF	Extruded anodised aluminium
Inputs, IMU, CAN, CAN Bridge, outputs	Identical in all versions (chap. 3)			

2.2 Lightweight version and rugged version (R)

The electronics and all technical characteristics are identical: only the housing differs.



Figure 1 – Lightweight version: PA12 + 35% GF housing.



Figure 2 – Rugged R version: extruded anodised aluminium.

- **Lightweight version**: housing in PA12 nylon filled with 35% glass fibre, a material with excellent mechanical properties, resistant to shocks and vibration. Ideal where weight is a key requirement.
- **R version**: extruded anodised aluminium housing, for maximum mechanical strength and long-term durability in harsh environments (high-temperature areas, inside machinery).

2.3 Plug & play CAN cable for motorcycles

GUBELOG connects to the CAN socket (EURO5) of many motorcycles with a dedicated wiring harness, which is **not included** in the package and is ordered separately (chap. 7). For these motorcycles DataStudio provides ready-made CAN configurations.



Figure 3 – GE-WR-0001 wiring harness for the EURO5 CAN socket.

Make	Models
Kawasaki	ZX-10R, ZX-6R 636 (2016–2026), Ninja 650 (2021–2026)
Yamaha	R1 / R1M, R6, R9 (2015–2026), R7 (2022–2026)
Honda	CBR 1000RR-R, CBR 600RR (2017–2026)
BMW	S1000RR (K67) (2019–2026)
Triumph	Street Triple / Daytona 765 (2017–2026)
MV Agusta	F3 800 (2018–2026)
Aprilia	RS660 (2020–2026)
Suzuki	GSX-8R (2024–2026)
Ducati	Panigale V4, V2 955, 899

3 Technical data

3.1 Channels and sampling

Group	Channels	Content
GPS	GPS1-GPS4	Latitude, longitude, speed, fix status
IMU	IMU1-IMU6	Accelerations X, Y, Z and angular rates X, Y, Z
DIGITAL	DIGITAL1-4	Frequency of the digital inputs DI1-DI4
ANALOG	ANALOG1-8	AN1-AN4 voltage, AN5-AN8 current
CAN	CAN1-CAN16	Signals extracted from the CAN bus

- Inputs recorded simultaneously: up to **38**.
- Sampling rate: from **0.01 Hz to 200 Hz**, programmable for each channel.
- Each channel has a label, unit of measurement, multiplier and offset, set from DataStudio.

3.2 GPS/GNSS receiver

	GUBELOG-01 / 01R	GUBELOG-03 / 03R
Update rate	1 Hz	10 Hz
Antenna	Integrated on the board	External (GPS module supplied)
Constellations	All GNSS constellations	
Recorded data	Latitude, longitude, ground speed, fix status	
Functions	Lap counting with GPS finish line (GPS trigger), auto-start on speed	

3.3 Inertial platform (IMU)

6-axis sensor (3-axis accelerometers and gyroscopes) rigidly mounted on the internal circuit board: it measures longitudinal and lateral accelerations and roll, pitch and yaw rates.

	Full scale	Internal sampling	Low-pass filter
Accelerometers (3 axes)	±30 g	1125 Hz	68.8 Hz
Gyroscopes (3 axes)	±2000 °/s	1125 Hz	73.3 Hz

3.4 Analog inputs (AN1-AN8)

	AN1-AN4	AN5-AN8
Type	Voltage 0-5 V, high impedance	Current 0-20 mA / 4-20 mA
Converter	12-bit ADC	
Limits	Permitted voltage from -0.3 V to 5.3 V	Internal 250 Ω precision resistor to ground
Typical sensors	Potentiometers, pressure sensors, pre-amplified thermocouples	Industrial current-output sensors, connected without additional hardware

The connector provides a 5 V output to power the sensors (chap. 3.13).

3.5 Digital / frequency inputs (DI1-DI4)

	Value
Applications	Wheel speed sensors, engine speed, flow meters, tachometric sensors
Frequency range	From 1 Hz to 20 kHz, high-frequency hardware counter (practically zero latency)
Logic thresholds	High level above 2.5 V, low level below 1.0 V
Maximum voltage	Supply voltage (VBatt)

3.6 CAN bus

	Value
Transceiver	High-Speed CAN 2.0B
Channels	Up to 16 simultaneous signals (CAN1-CAN16)
Bus speed	1 Mbps, 500 kbps, 250 kbps
ID format	11-bit (standard) or 29-bit (extended)
Protocols	OBD-II, SAE J1939, ISOBUS (ISO 11783), CANopen (EN 50325-4), CUSTOM signals (also from DBC files)
Termination	120 Ω resistor integrated in the data logger

3.7 Digital outputs and alarms

The logger manages 4 programmable alarms, each on one channel with a one- or two-threshold condition and hysteresis. Alarms 1 and 2 drive the two digital outputs DO1 and DO2, which generate a 0-5 V signal or drive two relays to activate external actuators (motors, solenoid valves, pumps).

	Value
Outputs	2 (DO1, DO2)
Signal	0-5 V
Relays	Drives externally powered relays
Maximum load	Max drain voltage 60 V, max current 1.5 A
Conditions	Channel > value 1, < value 1, between value 1 and value 2, outside value 1 and value 2

3.8 CAN Bridge

Available on all GUBELOG models. It creates a bridge between the physical inputs (GPS, IMU, digital, analog) and the CAN bus: the measured values are transmitted on the bus in real time, for example to a dashboard display or an ECU.

	Value
Transmitted groups	GPS LAT-LON, GPS SPEED-STATUS, IMU ACC, IMU GYRO, DIGITAL 1-2, DIGITAL 3-4, ANALOG 1-4, ANALOG 5-8
CAN ID	One per group, programmable
Frequency	OFF, 1, 10, 50, 100 Hz per group

> **CAN Bridge mode**
> In this mode the incoming CAN channels are not acquired and no data is recorded on the microSD.

3.9 WiFi module and cloud service (GUBELOG-01 / 01R)

The integrated WiFi module acts as a bridge to the local network and transmits the input values and the status of the 4 alarms in real time. Over the WiFi network the data can be sent to the **GUBECLOUD** service, which provides a monitoring dashboard and alarm notifications.

	Value
Network processor	WIZnet WizFi360 (Industrial Grade)
Standard	IEEE 802.11 b/g/n
Frequency	2.4 GHz (channels 1-13)
Protocols	Native TCP/IPv4 stack, real-time streaming on a programmable TCP port (default 9000)
Security	WPA / WPA2-PSK networks
Mode	Station (STA): automatic connection to the router or company network
Cloud upload interval	Off, 10 s, 60 s, 10 min, 60 min (instantaneous value or maximum of the interval)
Current consumption	Peak of about 200 mA when transmitting

External LTE module

> **Not yet available**
>
> The firmware and DataStudio provide for an external LTE module to send data to the cloud over the cellular network where no WiFi network is available. Planned parameters: APN, APN username and password, SIM PIN, upload interval (Off, 1, 10, 60, 120 min), cloud token, instantaneous value or maximum of the interval.

3.10 Memory (microSD)

	Value
Media	microSD in RAW format (unformatted); data is organised in datasets
Supported capacities	16, 32, 64 and 128 GB (higher capacities work but are not fully used)
Reading	Via USB-C or by removing the microSD and reading it on the PC in DataStudio
Recording start	*Always On* (at power-on) or *Auto-start* (when a condition on a channel occurs); the dataset is closed at power-off

3.11 USB-C

USB-C connector on the side of the housing, protected by a rubber cap. It is used to send and receive the configuration, monitor the active inputs in real time and download the datasets. Maximum transmission speed: 1 Mbps (FTDI virtual serial port).

3.12 Power supply

	Value
Supply voltage	From 7 to 24 V DC
Current consumption	About 80 mA (plus a peak of about 200 mA with WiFi transmitting, GUBELOG-01)
Power outputs	+5 V for sensors (pin 15), +12 V (pin 14)

3.13 Main connector JAE MX23A26NF1

All signals (analog, digital, CAN and power) pass through a single 26-way **JAE Electronics MX23A26NF1** automotive connector, designed for the car and motorcycle industry and for exposed areas (engine bay, external frames, agricultural machinery): silicone seals on the interface and on each wire against water, oil and dust, vibration-resistant snap latch (Click-Lock), physically separated pins against short circuits caused by moisture.

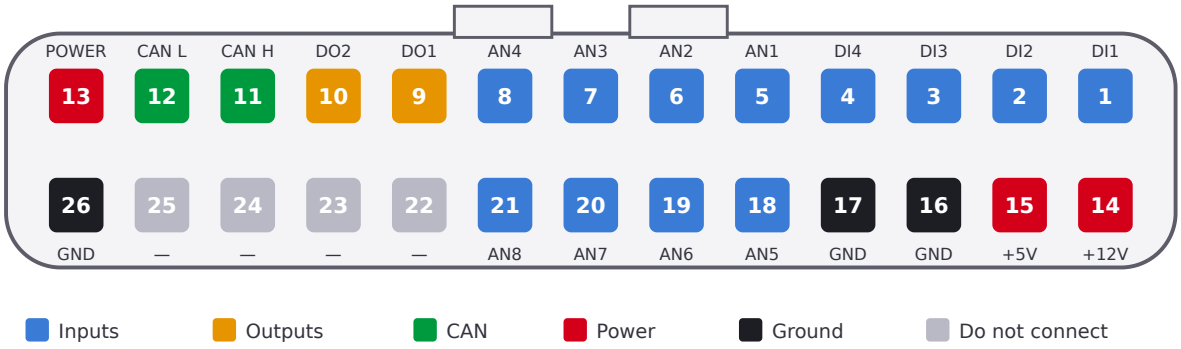


Figure 4 – Pin numbering of the logger connector, front view (as in the JAE manufacturer's drawing).

Pin	Signal	Type	Pin	Signal	Type
1	DIGITAL INPUT 1 (DI1)	Input	**14**	+12 V OUT	Power output
2	DIGITAL INPUT 2 (DI2)	Input	**15**	+5 V OUT (sensors)	Power output
3	DIGITAL INPUT 3 (DI3)	Input	**16**	GND	Ground
4	DIGITAL INPUT 4 (DI4)	Input	**17**	GND	Ground
5	ANALOG INPUT 1 (0-5 V)	Input	**18**	ANALOG INPUT 5 (0/4-20 mA)	Input
6	ANALOG INPUT 2 (0-5 V)	Input	**19**	ANALOG INPUT 6 (0/4-20 mA)	Input
7	ANALOG INPUT 3 (0-5 V)	Input	**20**	ANALOG INPUT 7 (0/4-20 mA)	Input
8	ANALOG INPUT 4 (0-5 V)	Input	**21**	ANALOG INPUT 8 (0/4-20 mA)	Input
9	DIGITAL OUTPUT 1 (DO1)	Output	**22**	Reserved (programming)	Do not connect
10	DIGITAL OUTPUT 2 (DO2)	Output	**23**	Reserved (programming)	Do not connect
11	CAN H	CAN bus	**24**	Reserved (programming)	Do not connect
12	CAN L	CAN bus	**25**	Reserved (programming)	Do not connect
13	POWER (7-24 V DC)	Power supply	**26**	GND	Ground

Pins 22-25

They are reserved for factory firmware programming: do not connect them.

3.14 Operating environment

	Value
Degree of protection	IP67
Operating temperature	From -20 °C to +85 °C

3.15 Dimensions and weight

	Lightweight version (01, 03)	R version (01R, 03R)
Housing	PA12 + 35% GF	Extruded anodised aluminium
Width × height	63 × 27 mm (+2.5 mm for the USB cap)	57 × 28 mm
Length (connector included)	95.3 mm (body 70.5 mm)	93.35 mm (body 68 mm)
Weight (logger only)	92 g	

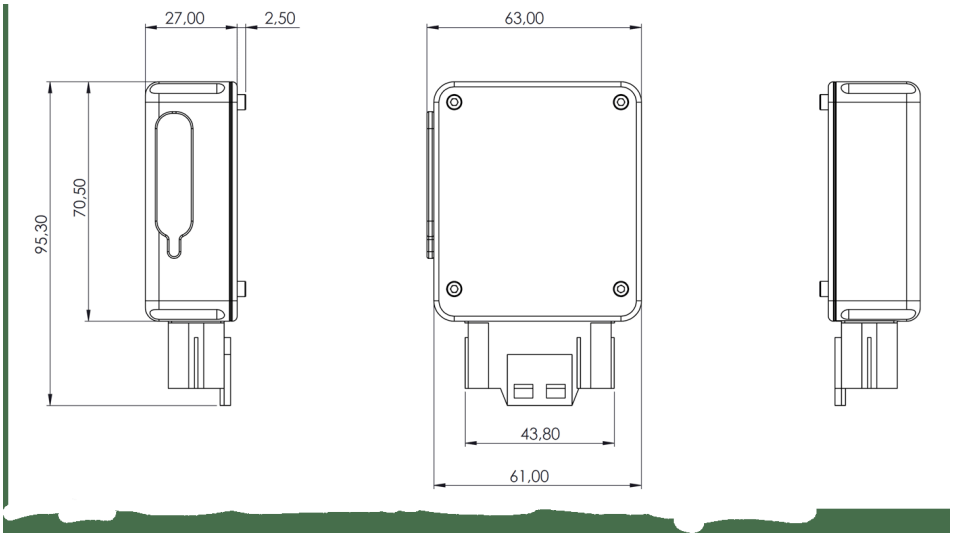


Figure 5 – Dimensions of the lightweight version (mm).

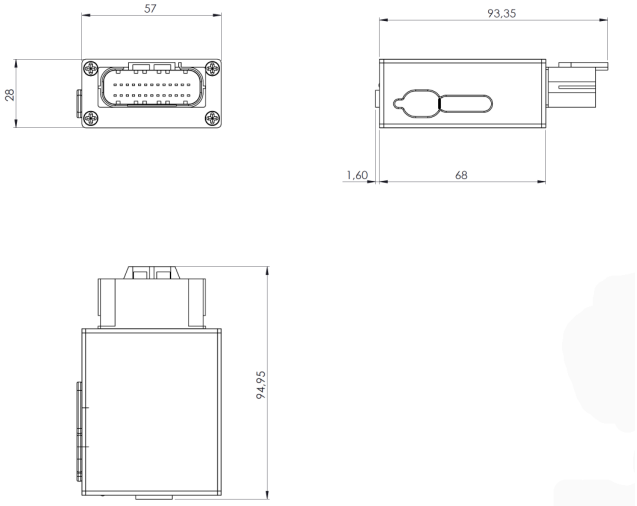


Figure 6 – Dimensions of the R version (mm).

4 Typical connections

Connection	Logger pins	Notes
Power supply	13 (+), 16/17/26 (GND)	From 7 to 24 V DC, for example from the vehicle battery via the ignition switch.
0-5 V sensor (e.g. potentiometer)	15 (+5 V), 5 (AN1), 16 (GND)	In DataStudio: multiplier and offset to convert volts into the sensor's unit.
Two-wire 4-20 mA sensor	14 (+12 V) or external supply, 18 (AN5)	The 250 Ω resistor is internal: 4-20 mA corresponds to 1-5 V.
Engine speed / wheel speed sensor	1 (DI1), 15 (+5 V), 16 (GND)	Pulse signal from 1 Hz to 20 kHz, high above 2.5 V.
CAN / OBD-II line	11 (CAN H), 12 (CAN L), 16 (GND)	120 Ω termination integrated in the logger.
Relay	9 (DO1), 10 (DO2)	Relay coil powered externally; max 60 V, 1.5 A.

Motorcycle EURO5 CAN socket (GE-WR-0001 wiring harness)

Motorcycle socket pin	Signal	Logger pin
16	Battery power	13
6	CAN High	11
14	CAN Low	12
4	Chassis ground	26
5	Signal ground	16

5 Configuration with GUBELLINI DataStudio

5.1 The software

Software	What it does
GUBELLINI DataStudio (free)	Configures Gubellini Electronics devices (GUBELOG, MICROLOG, GUBENAV, CAN boards); imports signals from DBC files; downloads datasets via USB or from the microSD and exports to MoTeC, AEM and CSV; shows values in real time on a dashboard; receives data via WiFi and forwards it to the cloud; manages datasheets and part counters (hours and km); includes an AI assistant for sensor connection, configuration and diagnostics.
GUBELLINI DataAnalysis (free)	Analyses the downloaded datasets: plots versus time or distance, layouts with multiple plots, math channels, track map, import and export of on-board video; it includes an **AI assistant** that analyses the data, writes reports and generates the code of math channels. Designed for motorsport and automotive, suitable for any application.

The DataStudio user manual, included in the program (**Help → User manual (PDF)**), describes every function in detail; the main steps for GUBELOG are summarised here.

5.2 First connection

1. Install DataStudio and connect the powered GUBELOG to the PC with the USB-C cable. Windows installs the FTDI virtual serial port driver.
2. In the main window press **GUBELOG**: in the status bar, under **Port**, the logger's COM port is proposed.
3. **Logger → Connect / information** shows the model and firmware; **Read configuration** loads the logger's configuration.

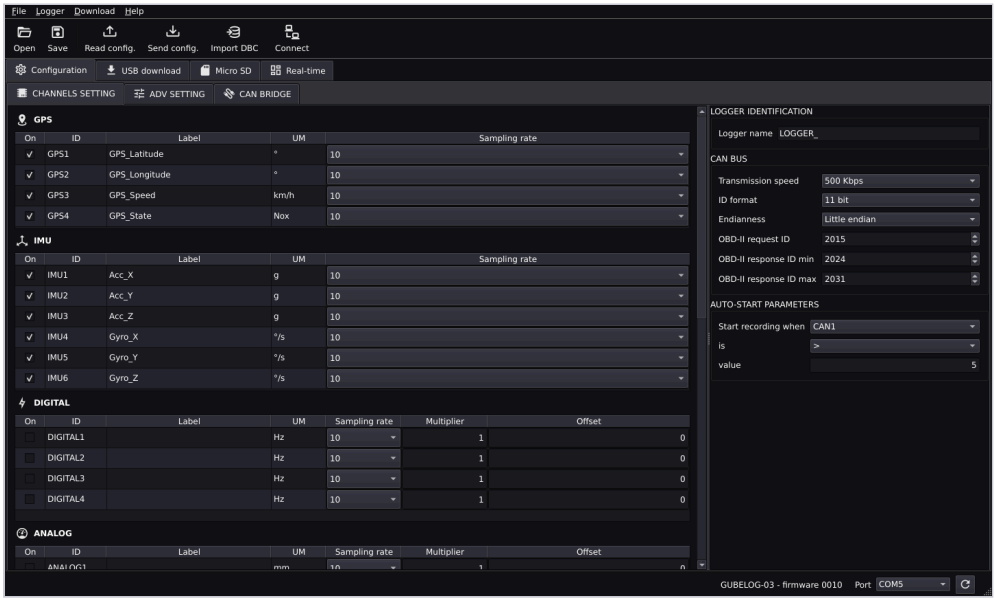


Figure 7 – The GUBELOG configuration in DataStudio.

5.3 Channels, labels and frequencies

In the **CHANNELS SETTING** tab you enable each channel to be recorded, give it a name (label) and a unit of measurement and choose the frequency. For digital, analog and CAN inputs you also set the **multiplier** and **offset**: value = raw × multiplier + offset.

Example: 4-20 mA pressure sensor, 0-10 bar on AN5

The logger measures 1-5 V (250 Ω resistor). 1 V corresponds to 0 bar, 5 V to 10 bar: multiplier = 10 / 4 = 2.5 bar/V, offset = -2.5 bar.

Recording time

The logger records at the frequency of the fastest channel: choose for each channel the lowest useful frequency (chap. 6).

5.4 Digital frequency inputs

The DIGITAL channels record the signal frequency in Hz. With the multiplier it is converted into the desired quantity.

Example: engine speed with 1 pulse per revolution

rpm = Hz × 60: multiplier 60, offset 0. With a 36-tooth tone wheel for wheel speed: wheel revolutions per second = Hz / 36; for km/h also multiply by the circumference in metres and by 3.6.

5.5 CAN channels

- **CUSTOM**: for each channel you set the CAN ID, start bit position, length, type (signed or unsigned) and format (Intel/Motorola), then multiplier and offset.
- **OBD-II and J1939**: predefined signals (e.g. engine speed, speed, throttle) chosen from a list.
- **DBC import**: **File → Import DBC…** fills in the CAN channels from the signals of a DBC file.
- **Motorcycles**: for the motorcycles in § 2.3 ready-made configurations are available, to be opened with **File → Open…**.

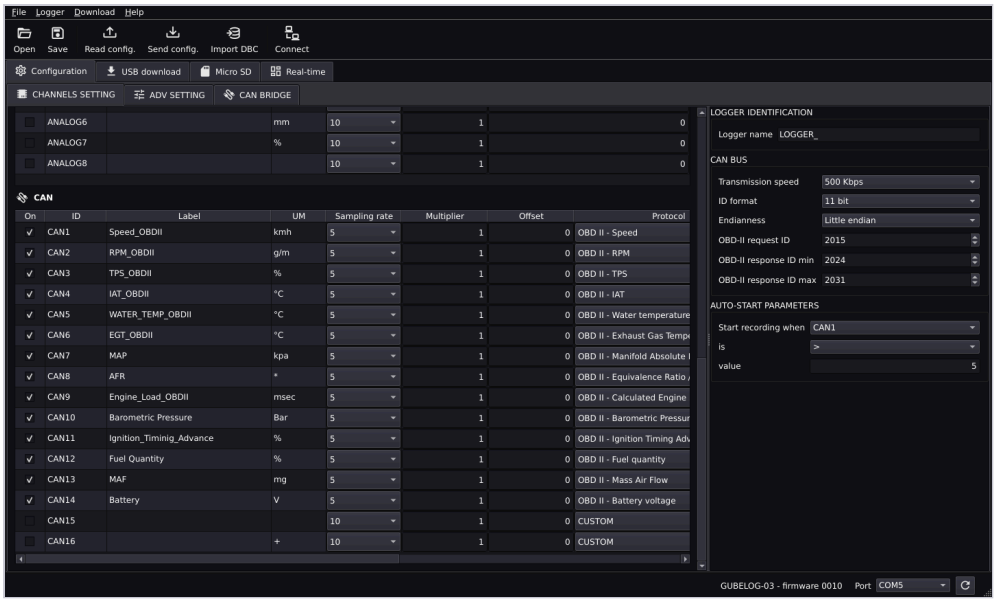


Figure 8 – The CAN channels in DataStudio.

5.6 Auto-start conditions

In **AUTO-START PARAMETERS** you choose *ALWAYS ON* (recording from power-on) or a channel, a condition (=, >, <) and a threshold: the dataset is created when the condition becomes true.

> **Example: record only when moving**
> Start recording when **GPS3** (GPS speed) **> 10** km/h.

5.7 GPS trigger and track list

In **ADV SETTING → GPS** you set the latitude and longitude of the finish line and the radius of the area (from 5 to 100 m): the logger counts a lap at each crossing. In Motorsport and Automotive mode the **Track** drop-down list offers about 300 circuits from all over the world and fills in the coordinates and recommended radius.

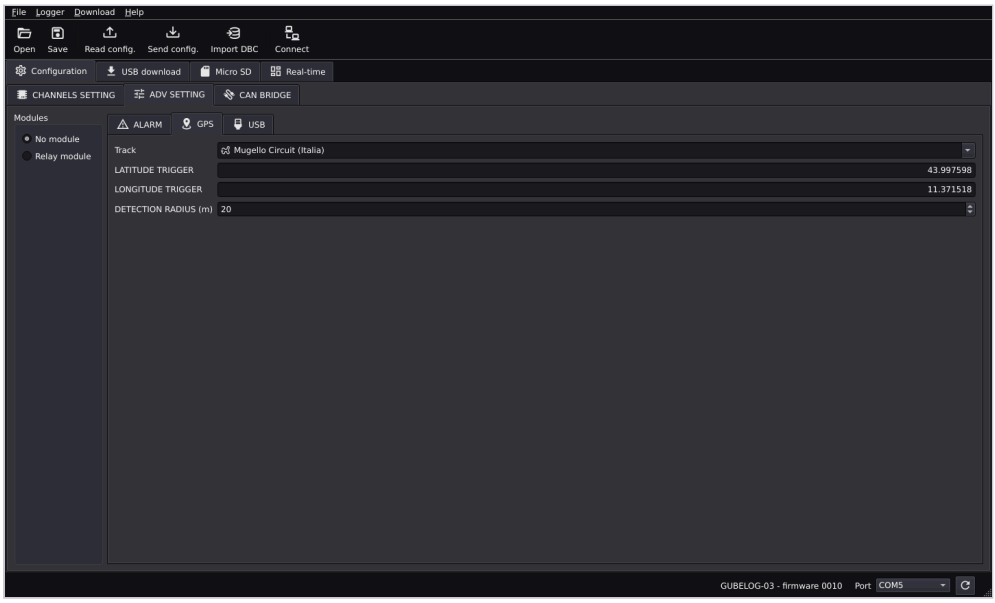


Figure 9 – The GPS finish line with the choice of circuit.

5.8 Alarms and digital outputs

In **ADV SETTING → ALARM** you configure the 4 alarms: channel, condition, one or two thresholds and hysteresis. Alarms 1 and 2 drive outputs DO1 and DO2; the status of the 4 alarms is also transmitted via WiFi.

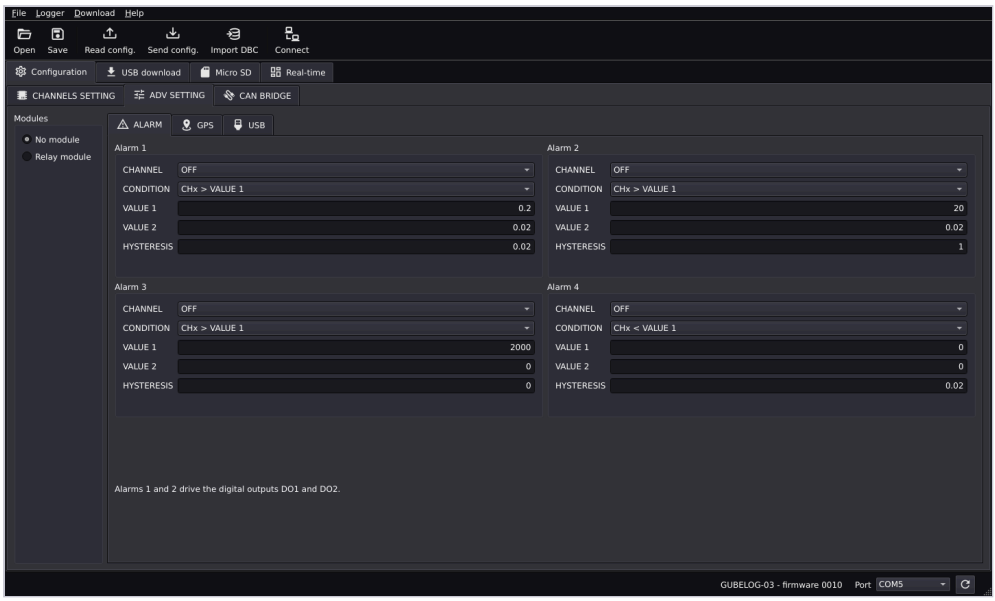


Figure 10 – Alarm configuration.

5.9 WiFi, WiFi Monitor and cloud (GUBELOG-01)

In **Industrial** usage mode, in the **ADV SETTING → WIFI** tab you enter the network (SSID), password, TCP port, interval and cloud token. The DataStudio **WiFi Monitor** module shows the received values and the alarm status in real time, and can forward them to the cloud.

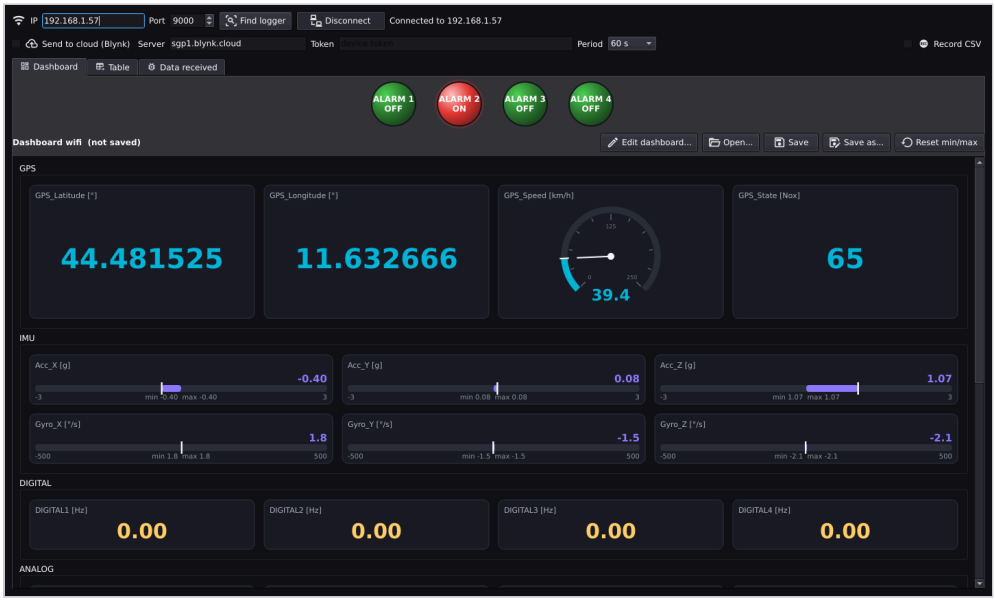


Figure 11 – The DataStudio WiFi Monitor.

5.10 CAN Bridge

In the **CAN BRIDGE** tab you choose, for each group of inputs, the CAN ID and the transmission frequency. Remember that in this mode no data is recorded.

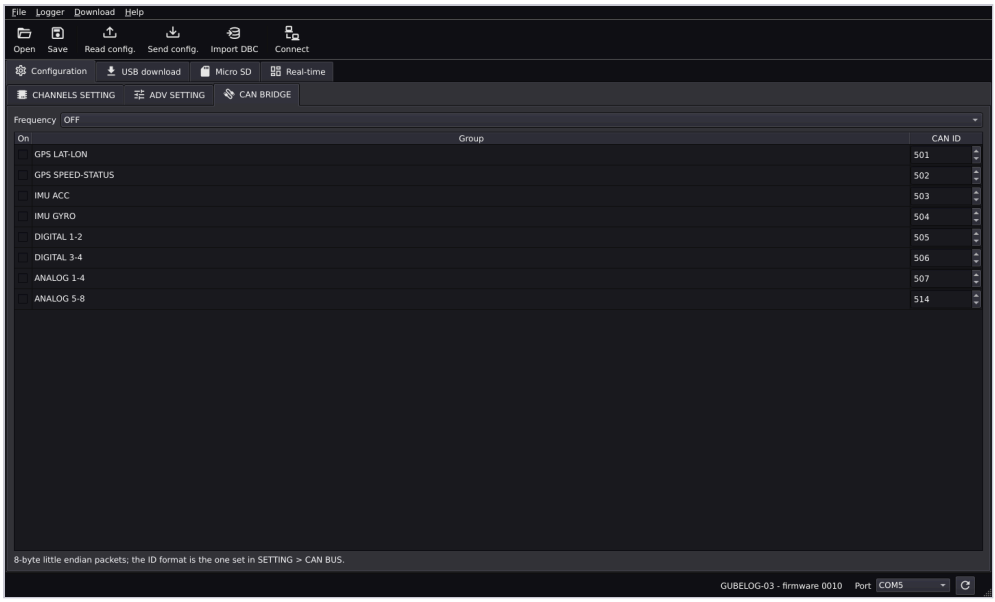


Figure 12 – The CAN BRIDGE tab.

5.11 Sending, download and analysis

1. **Logger → Send configuration**: the logger saves the configuration and restarts. DataStudio assigns an ID to the configuration, so that each dataset gets the right labels.
2. After the tests, download the datasets from the **USB download** tab or by reading the microSD from the **Micro SD** tab.
3. Open the `.dat` file with GUBELLINI DataAnalysis (**Analysis** button in the main window).

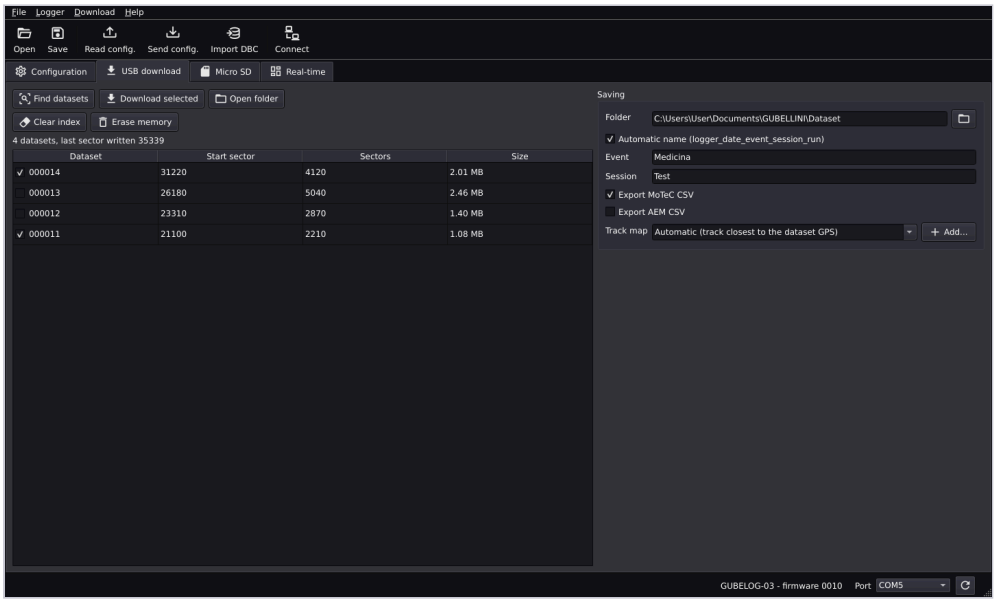


Figure 13 – Downloading datasets via USB.

5.12 AI assistant

The DataStudio **AI assistant** answers questions about connection and configuration, checks the open configuration (for example disabled auto-start channels, frequencies, CAN parameters) and proposes changes, which the user can apply or reject; with Real-time running it performs sensor diagnostics. It uses a personal API key from an AI provider.

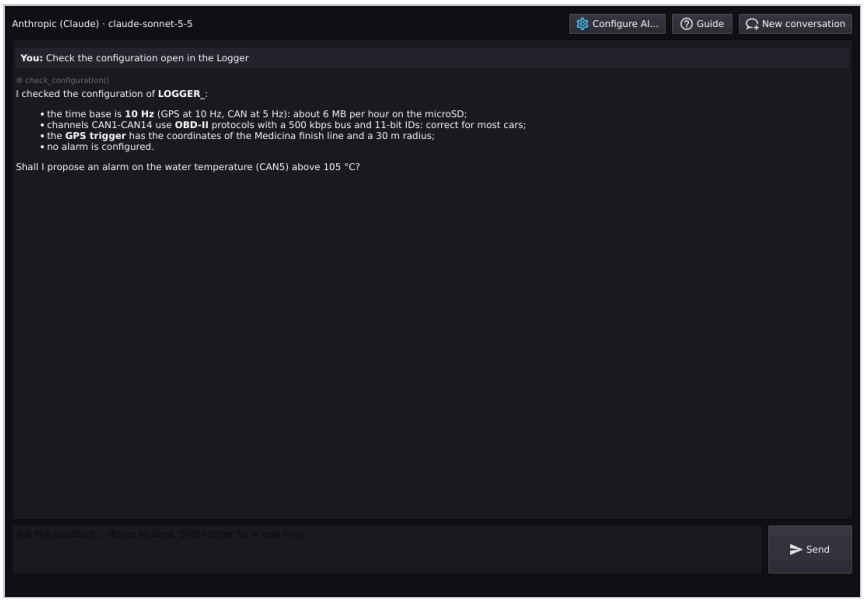


Figure 14 – A question to the AI assistant.

6 microSD recording time

The logger records all channels at the frequency of the fastest active channel; each 512-byte sector contains 3 complete samples, regardless of the number of active channels.

Recording frequency	Data per hour	16 GB microSD	32 GB microSD	128 GB microSD
1 Hz	about 0.6 MB	over 25,000 hours	over 50,000 hours	over 200,000 hours
10 Hz	about 6 MB	about 2,600 hours	about 5,200 hours	about 20,800 hours
50 Hz	about 31 MB	about 520 hours	about 1,040 hours	about 4,160 hours
100 Hz	about 61 MB	about 260 hours	about 520 hours	about 2,080 hours
200 Hz	about 123 MB	about 130 hours	about 260 hours	about 1,040 hours

7 Ordering information

7.1 Data loggers

Code	Description
GE-GL-0001	GUBELOG-01 – PA12 + 35% GF housing, WiFi module
GE-GL-0001R	GUBELOG-01R – aluminium housing, WiFi module
GE-GL-0003	GUBELOG-03 – PA12 + 35% GF housing, 10 Hz GPS
GE-GL-0003R	GUBELOG-03R – aluminium housing, 10 Hz GPS
GE-GL-KT03R	GUBELOG-03R kit for EURO5 motorcycles: logger, external GPS module, GE-WR-0001 wiring harness

Package contents: logger, external GPS module (GUBELOG-03 / 03R), USB-C cable, instruction manual, GUBELLINI DataStudio and GUBELLINI DataAnalysis software.

7.2 Wiring harnesses

Code	Description
GE-WR-0001	60 cm wiring harness to connect the logger to the EURO5 CAN socket
GE-WR-0002	60 cm wiring harness with EURO5 CAN socket and 2 connectors for 0-5 V analog sensors

7.3 Sensors

Code	Description
GE-SK-0001	Suspension potentiometer kit: 1 × 150 mm and 1 × 75 mm
GE-LP-150	Linear suspension potentiometer, 150 mm
GE-LP-75	Linear suspension potentiometer, 75 mm
GE-TCK-30500-M	Mineral-insulated type K thermocouple
GE-PRS-16B	Pressure sensor 0-16 bar (1/4" BSP thread)
GE-PRS-40B	Pressure sensor 0-40 bar (1/4" BSP thread)
GE-NTC-M8-10K	NTC 10K temperature sensor, M8×1 thread
GE-PROX-M8-3W-NO	M8 inductive proximity sensor (3-wire DC, N.O.)
GP-GS-TAS514-100K	Miniature tension and compression load cell, 100 kg

7.4 Related products

Code	Description
GE-GN-0001	GUBENAV – high-dynamics 100 Hz GPS and IMU sensor, on CAN
GE-CB-0001…0005	CAN-BOARD: accelerometer, GNSS, load cell, OPTO, type K thermocouple – values transmitted on the CAN bus
GE-ML-0001	MICROLOG-01 – CAN logger with 16 CAN channels

Information and orders
GUBELLINI s.a.s. di Diego Gubellini & C. – Via Euridia Bergianti 10B, 40059 Medicina (BO) Italy
E-mail: info@gubellinielectronics.com · Tel. +39 347 9200208 · WhatsApp +39 380 1978288
www.gubellinielectronics.com

Technical data are subject to change without notice.